07 | Endpoint and Path IG: state, seed and equality

Compare the same target and input coordinates before comparing images.

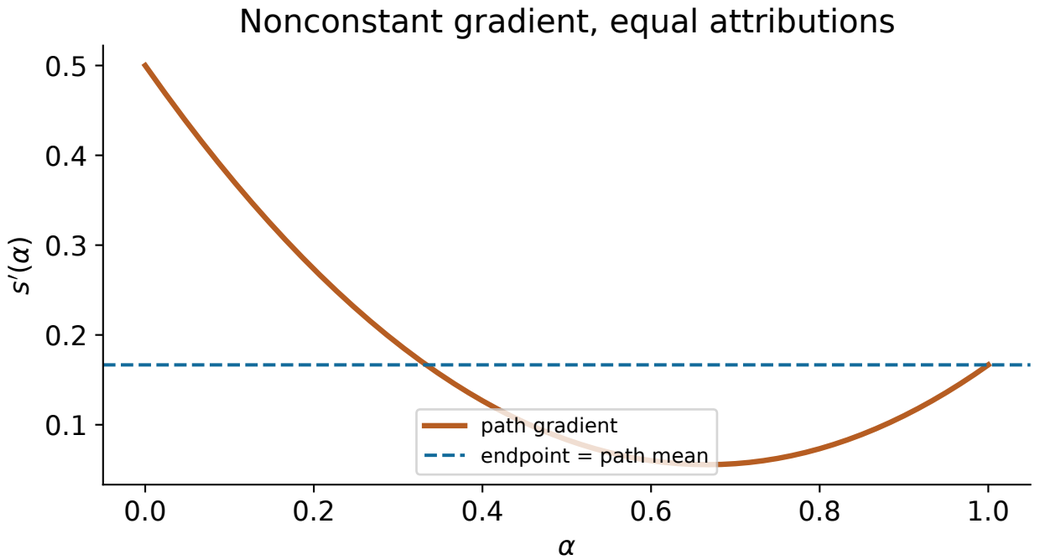
$$L^{end} = x \odot \nabla s(x)$$

$$L^{\Delta end} = \Delta x \odot \nabla s(x)$$

$$IG = \Delta x \odot \int_0^1 \nabla s(x_\alpha) d\alpha$$

$$L_i^{\Delta end} = IG_i \iff \Delta x_i \left[\partial_i s(x) - \int_0^1 \partial_i s(x_\alpha) d\alpha \right] = 0$$

Constant path Jacobian is sufficient, not necessary. Canonical Endpoint also has a different seed if x0 is nonzero.



$$s(t) = t^3/3 - 2t^2/3 + t/2$$

$$x_0 = 0,\ x = 1: \quad L^{end} = IG = 1/6$$

$$x = (r - \mu)/\sigma$$

$$x_i \partial_{x_i} s = (r_i - \mu_i) \partial_{r_i} s$$

Normalized-input endpoint depends on the coordinate origin; it does not require a baseline forward or imply completeness.